Plant Care Scheduler – Application Reworking Plan

1. Project Objective

Improve the existing Plant Care Scheduler application by refining its modules, introducing the agreed features, and maintaining a professional, user-friendly interface.

Target: Complete the agreed application rework within one week.

2. Module-wise Requirements

1. Dashboard

Status: Changes required

- Remove the duplicate application name appearing in the logo and heading.
- Replace the excessive dark-green theme with a balanced colour palette.
- Display useful information such as plant count, upcoming care tasks, completed tasks, and weather information.

2. User Authentication

Status: No changes to be made

3. My Plants

Status: Changes required

- During plant registration, ask for the plant's placement:
    1. Indoor
    2. Outdoor – Open area
    3. Outdoor – Under shelter
- Use the geographical location stored in the user's Profile instead of collecting location separately for each plant.

4. Care Tasks

Status: Changes required

- Introduce a calendar to view scheduled, completed, and missed tasks.
- Completed tasks should disappear from the active task list but remain accessible in calendar/history.

- Integrate weather-aware watering recommendations for outdoor plants.

5. Environmental Data (Previously Sensor Module)

Status: Changes required

- Retain this module for weather and environmental information.
- Integrate a weather API using the geographical location stored in the user's Profile.
- Retrieve and store relevant information such as temperature, humidity, and rainfall/forecast data.
- Use environmental information to support weather-aware plant-care recommendations.
- Check weather conditions when a watering task is approaching.
- For outdoor plants in open areas, notify users when rainfall is expected and suggest reconsidering watering.
- For sheltered outdoor plants, consider that rainfall may not reach them directly.
- Ignore rainfall-based recommendations for indoor plants.
- Do not automatically cancel watering tasks; let users decide.

7. Health Records

Status: Changes required

- Automatically capture the assessment date and time.
- Provide an optional next follow-up date/time.
- Maintain the plant's health history.

8. Community

Status: Changes required

- Support uploading photos along with posts.
- Display community posts and their associated images.

9. Plant Species Guide

Status: No changes to be made

10. Plant Care Chatbot

Status: New functionality to be implemented

- Integrate a chatbot for plant-care-related questions.
- Support questions about watering, sunlight, plant diseases, and general gardening guidance.
- Make the chatbot accessible within the application.

11. User Profile

Status: Changes required

- Store the user's geographical location here.
- Store latitude and longitude if required for accurate weather lookup.
- Use this location for Environmental Data retrieval and weather-aware recommendations.

12. Notifications

Status: Changes required

- Provide notifications for upcoming care tasks and reminders.
- Notify users about relevant weather-based watering recommendations.
- Avoid unnecessary repeated notifications.

13. Consultation

Status: Remove completely

- Remove the Consultation module.
- Remove the Consultation entity and its relationships from the ER diagram.
- Remove its related backend and frontend functionality.

5. One-Week Reworking Schedule

Day	Main focus
Day 1	Review existing code, finalize ER diagram, and update database design.

Day 2	Dashboard redesign, Profile, and My Plants improvements.
Day 3	Care Tasks, calendar, completion history, and reminders.
Day 4	Environmental Data and weather-aware care recommendations.
Day 5	Health Records and Community improvements, including image uploads.
Day 6	Chatbot implementation, Authentication, and Spring Security.
Day 7	Notifications, testing, bug fixing, and final UI refinements.

Prioritize the agreed core functionality within the one-week target. Any unfinished advanced functionality can be taken up in a later phase.

6. Development Guidelines

- Work with the existing application; do not rebuild modules unnecessarily.
- Follow the module-wise status and requirements listed above.
- Do not introduce features beyond the agreed scope.
- Review existing implementation before modifying a module.
- Maintain a consistent, professional, and responsive UI.
- Preserve existing functionality wherever no changes are required.
- Complete and review the ER diagram before implementing database changes.